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Spatial Diffusion of E-Commerce in China's Counties: Based on the Perspective of Regional Inequality

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Abstract: In recent decades, China has been on a new journey toward a digital economy of which e-commerce accounts for a substantial proportion. Despite some controversy, the innovation diffusion hypothesis and efficiency hypothesis of online shopping have been tested in research on the urban–rural dual structure. However, research on the spatial diffusion model of online business is sparse. Based on the online business and online shopping index released by the Ali Research Institute, this article compares the spatial diffusion model of online shopping and online business in the core–periphery structure based on the inequality between the eastern and western regions of China. Our study suggests that online business trends are in line only with the innovation diffusion hypothesis, with marginal counties having lower levels of online business. Online shopping, on the other hand, is in line with the innovation diffusion hypothesis and partially with the efficiency hypothesis, with a higher index of online shopping in the core regions and some peripheral counties. The discrepancy in the spatial diffusion mode is due to the differences in aims and supporting elements between online business and online shopping. Apart from infrastructure, the diffusion of online business is largely constrained by the regional industrial base, while online shopping is influenced by income and savings levels, which is the main reason for the differences in the spatial diffusion of online business and online shopping. We argue that the diffusion of online business has not led to the ability to balance regional inequalities at the national scale, while online shopping has the potential to bridge core and peripheral disparities better than online businesses, not in terms of the ability to bridge economic disparities, but in terms of the potential to reduce spatial consumption inequalities and welfare gains.

Keywords: online business; online shopping; spatial diffusion model; innovation diffusion hypothesis; efficiency hypothesis; regional inequality

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1. Introduction

Since the advent of information and communication technologies (ICTs), there have been ongoing debates about the spatio-temporal effects of the spatial expansion of the internet. The impact of the spread of the internet on the geographic location of economic activity has also been the subject of much scholarly attention [1,2]. In response to the technical characteristics of the internet, the theory of "time-space compression" was proposed [3]. Moreover, Cairncross [4] declared the "death of distance", and Friedman [5] stated that "the world is flat" in the internet era. It is indisputable that the application of the internet overcomes the friction caused by distance and reduces the communication, transportation and search costs for economic activities [6], and the development of networks creates decentralization in the core–periphery structure.

However, different views on the impact of the internet remain. In the urban–rural dichotomy, the central city is considered to be an important distribution area for internet technologies. Compared with urban areas, rural areas are obviously high-cost markets.

Thus, investment in new technologies will continue to be made in urban regions, and this urban orientation has led to a gap in digital diffusion between urban and rural regions [7]. Furthermore, differences in social and economic characteristics, such as knowledge, skills, and income, have also led to social exclusion in the adoption of internet technologies [8]. Thus, the polarizing trend of the core area within the core–periphery structure of urban and rural regions has also been repeatedly emphasized in the adoption of the internet.

Research on the spatial expansion of the internet between urban and rural regions has focused on the commercial application of the internet—e-commerce—as a breakthrough. Anderson et al. [1] proposed two mutually targeted hypotheses regarding whether the adoption of online business might affect the spatial structure, namely, the innovation diffusion hypothesis, based on the innovation hypothesis, and the efficiency hypothesis, based on the demand hypothesis. Compared with residents in other regions, people living in urban regions tend to use e-commerce because of their higher education and internet skills [9]. However, the adoption of online shopping in remote areas and sparsely populated areas lacking stores is particularly valuable in overcoming spatial friction [2]. There has been an interesting debate as to whether e-commerce usage in rural areas will exceed that in urban areas [6].

Previous testing of the spatial diffusion hypothesis of e-commerce has focused on the municipal level. Two of Anderson's hypotheses on the spatial diffusion of online shopping are tested by comparing the differences in online shopping adoption between urban areas and suburban or rural regions. It is also common in research to equate e-commerce directly with online business or online shopping in order to study the differences between urban and rural areas, without clearly distinguishing between online business and online shopping. Our study contributes to the growing body of research on the diffusion of e-commerce by providing a different perspective for understanding the expansion of e-commerce at the national level. For example, we distinguish between online shopping and online business and extend the core–periphery structure from the urban scale to the regional scale.

This article takes mainland China as the research scale and counties, except for municipal districts, as the research unit. Compared with the city unit, the number of county units is greater, allowing the precise identification of more detailed features of diffusion. We adopt the core–periphery structure based on the regional inequality between the eastern and western regions of China. The eastern part of China is economically developed and technologically advanced, while the remote areas in the west have low population and economic density. There is a gap in technological adaptability between the western and eastern regions. Therefore, we adopt the core–periphery structure with the east as the core area and the west as the marginal area based on the socioeconomic gradients of different regions in China. Previous tests of the innovation diffusion hypothesis and efficiency hypothesis have mainly been based on the diffusion of online shopping, using regression methods to analyze the econometric correlation between online shopping and urban or rural areas as a way of testing the diffusion hypothesis. Starting from the two opposites of "retail" and "consumption" in e-commerce, this paper tests and compares the applicability of online retail and online shopping in the innovation diffusion hypothesis and efficiency hypothesis. In this process, we discard the regression method and return to the essence of diffusion by using the diffusion gradient layout characteristics to determine which hypothesis best explains the large-scale diffusion of e-commerce. The mechanisms underlying the differences in the spatial diffusion patterns between online business and online shopping are also analyzed.

The following section introduces the literature on the diffusion of the internet and the spatial diffusion of e-commerce. Our review focuses on the digital divide caused by internet diffusion, the mechanism of internet diffusion, and the spatial diffusion of e-commerce. Section 3 elaborates on the data sources. Section 4 compares the spatial diffusion model of online business and online shopping. Section 5 uses the ordinary least squares (OLS) method to analyze the diffusion mechanism of online shopping and online business in different samples. Conclusions are drawn and discussed in Section 6.

2. Literature Review

2.1. Internet Diffusion and the Spatial Digital Divide

The digital divide between regions plays a considerable role in the proliferation of e-commerce, and in particular, the layout of digital infrastructure can significantly influence the adopting of e-commerce in different regions. The term "digital divide" is used to describe the difference between the "haves" and "have-nots" of digital infrastructure, referring to the wide social and regional differences in people's capacity to access and adopt digital equipment and services as well as their capacity to access the internet in terms of physical connectivity and use of facilities [10,11]. OECD (2001) described the "digital divide" as the gap between groups of different socioeconomic levels in terms of access to ICTs and the use of the internet for various activities. The digital divide has also been described as disparities in internet access, usage, search strategies, social support, information evaluation, and user diversity [12].

To conceptualize the digital divide, the dichotomy of the digital divide between different scales was a valuable concept at the beginning of the development of the internet, and most of the policy discussions and research on telecommunications issues have focused on the issue of access to technical facilities [13–15]. However, as the digital infrastructure divide has narrowed, research in recent years has diversified this binary concept, and new intricacies have been recognized in internet-access disparities [16,17]. Access to ICT is not the same as adoption or, more importantly, effective use. As a result, statements such as redefining the digital divide and transcending access rights have emerged [18,19]. Numerous empirical studies have emphasized the importance of distinguishing between access and use, as having access to an internet infrastructure does not mean that people will use it [20]. The redefinition of the concept of the digital divide focuses on the inequality of internet use, which is referred to as the deepening digital divide [21,22]. Following discussions on adoption, the issue of digital usability has become the third level of the digital divide, where dissimilar skill levels lead to new disparities [23]. This third level of the digital divide is related to the knowledge gap, and therefore, a change in the concept of the digital divide is necessary. The digital divide is not only a technological issue, but also a social issue that reflects broader social, economic, cultural, and learning inequalities [24].

Apart from the conceptual issues of the digital divide, research on the inter-regional digital divide is more oriented toward spatial scale studies and different objects. The digital divide coexists in different dimensions and forms, including digital divisions at different spatial scales and with different socio-demographic characteristics, and they are also linked to the ownership of ICT. Unlike the divide that stems from technology, people in marginalized areas are also affected by their own social exclusion factors, such as geographic location and social characteristics. The existing social divide needs to be reproduced in studies of digital inequality, which is a more pressing issue than the digital infrastructure gap in marginalized areas.

Digital inequalities involving spatial scales, including between countries, between urban and rural areas, and within cities and villages, have been explored by scholars. Early studies of inequality on the internet dealt with the inequalities in connectivity between developed and developing countries, and how these differences might affect society [10]. Compared to individuals in suburban and urban areas, studies have found that rural residents are less likely to use internet technology for economic and other everyday activities, a relationship that is a product of the slow diffusion of advanced technology to rural areas [25]. ICT adopters can be categorized according to their stage of adoption, and there is evidence that marginal areas will continue to lag behind as technologies continue to iterate [26,27]. Not only do marginal communities have less broadband internet usage than metropolitan communities, but also the degree of broadband availability varies within marginal communities [28].

Factors contributing to differences in ICT systems, classified according to socio-demographic characteristics, are socio-economic status in relation to the state of the technological infrastructure, including age, income, highest level of education attained,

ethnicity, etc. Looking back at rural issues a decade ago, low levels of education were a major barrier to ICT development in marginal areas [29]. Nowadays, education remains a huge factor in the internet use divide in marginalized communities [30]. For people living in marginal areas, rural communities may not be able to afford superfast broadband connections on their incomes, and even regular broadband can be no small challenge [31,32]. There is also a digital divide between the sexes, with the act of sharing online being very different in terms of gender, with men more likely to be involved [33,34]. Social factors affecting the digital divide also include age, with older people generally having lower access to ICT [35,36], and persistent gaps between different racial and ethnic groups [37].

2.2. Mechanism of Internet Diffusion

The diffusion of innovation is a long-term and continuous process, during which the share of companies adopting and applying new technologies grows [38]. Hägerstrand considers distance to be the main influencing factor for the diffusion of innovation, especially distance from innovation centers [39]. A number of empirical studies have proven that distance is negatively related to the degree of innovation diffusion [40,41]. Spatial proximity, cognitive proximity, social proximity, organizational proximity and institutional proximity have also received extensive attention from geographers in the study of innovation diffusion [42–44].

In terms of the diffusion mechanism, Hargittai summarized that the economic level, human capital, institutional environment and existing technology accumulation are all important elements in the diffusion of the internet in a country or city [45]. The spatial diffusion of the internet in the United States and the European Union is based on the theory of neoliberal economics [46]. Notably, the diffusion of internet access is a process driven by the market and profit, and cities always take the lead in internet diffusion [47]. The spatial diffusion of digital networks needs to create value for all major stakeholders. In particular, only when all participants can obtain appropriate value in the form of economy, efficiency, legitimacy, knowledge or social benefits can the network be successfully spread [48]. Guillen and Suarez conducted a survey using national-level data and demonstrated that internet adoption is a complex phenomenon and that the spatial diffusion of the internet is more influenced by per capita income, infrastructure, English language proficiency and entrepreneurial conditions than by the impact of public policy on telecommunications [49]. Additionally, Beilock and Dimitrova identified that per capita GDP is the most important factor affecting the rate of internet adoption [50]. According to the research of Lin et al., the proliferation of the internet is positively correlated with economic growth, with spillover effects being more pronounced in developed regions [51].

2.3. Spatial Diffusion of E-Commerce

From the perspective of "retail" and "consumption", the combination of the internet and commerce has resulted in two types of technology applications: business-oriented online business and consumer-oriented online shopping, which are collectively referred to as e-commerce [1]. Research on the spatial diffusion of e-commerce has focused on comparing the diffusion order and diffusion mode between core and marginal areas.

Most of the research on the spatial diffusion of online shopping has aimed to verify the two mutually targeted hypotheses proposed by Anderson. Farag et al., for example, indicated the spatial distribution of internet users and online buyers in the Netherlands from 1996 to 2001 and the impact of spatial variables on online shopping [2]. They found that the innovation diffusion hypothesis and the efficiency hypothesis coexist in the diffusion of e-commerce. On the one hand, residents living in highly urbanized regions are more likely to search or purchase online. On the other hand, residents living in less urbanized or non-urbanized areas with lower accessibility to stores can purchase products online more often. As De Blasio noted, the internet cannot reduce the role of distance, with urban consumers using the internet and e-commerce much more frequently than consumers in weakly urbanized areas [52]. According to the survey of Motte-Baumvol et al., the efficiency

hypothesis is partially validated by findings that suburban and urban households have very different patterns and that online purchases can remove spatial constraints on access to tangible goods [53]. Cao et al. examined the relationship between spatial attributes and online shopping and found supporting evidence for the innovation diffusion hypothesis, arguing that internet users living in cities or areas with high shopping convenience are inclined to shop online more frequently than those in other regions, as the former are more educated and use the internet more than the latter [54]. Beckers et al. highlighted the fact that there is no significant effect of the level of urbanization of a region on the probability of online shopping by analyzing data from a survey conducted by the Belgian Retail Federation with over 1500 respondents [55].

Research on the spatial diffusion hypothesis of e-commerce has long been centered on online shopping. However, does the spread of online business follow the innovation diffusion hypothesis and efficiency hypothesis? Do residents in marginal areas adopt the internet in order to overcome the distance barrier to sell their agricultural products? Previous studies of the spatial diffusion of online business included two main issues: the extent to which the diffusion of online business affects the number of visits to offline stores, and the comparison of the order of diffusion in urban and rural areas [6]. We focus primarily on the latter question. Retailers in urban areas are more likely to take on an online sales strategy, as urban areas typically have more advanced telecommunications infrastructures, stronger internet innovation capabilities and knowledge spillover, and more internet-related production services [56]. Clarke et al., for example, investigated a large-scale commercial consumer survey in the U.K. to explore the expansion of e-commerce in British retail and to examine the spatial differences in e-commerce adoption [6]. According to their research, e-commerce initially spread from London and major cities but was not just an urban trend. Due to the improvement of broadband service quality, usage in rural areas is increasing.

2.4. Digital Inequality and Spatial Diffusion of E-Commerce in China

The distribution of internet users in China is characterized by significant spatial differences, with the eastern coastal regions accounting for more than 50% of the country's internet users. China is the world's largest ICT market in terms of the number of mobile devices used, the number of internet users and the number of broadband users, but there are significant numerical inequalities within and between provinces, municipalities and counties [57]. The growth of e-commerce in China exhibits national inequalities that are constrained by local economic, political and infrastructural conditions. The growth of e-commerce exhibits a spatial mix that relies on the structure of cyberspace and physical space [58]. As a rural e-commerce cluster with Chinese characteristics, there are many studies on the spatial distribution of Taobao villages that involve the spatial diffusion of e-commerce. Similar to the regional development of the ICT industry, Taobao Villages first emerged in more developed regions and then gradually spread to less developed regions, showing a tendency of agglomerative diffusion [59]. As Liu et al. noted, more than 90% of Taobao villages are located in East China, while Central China has the second highest number of Taobao villages and the least in the West [60]. The study by Geng et al. also shows that, overall, the concentration of e-commerce in the northern and western regions of China is decreasing, with a stable trend of geographical concentration of e-commerce activity in the eastern and southern regions [61]. The use of e-shopping technology is also very unevenly distributed across regions, with more developed cities in the east being higher than those in the less developed west, and cities at higher administrative levels being higher than those at lower administrative levels [62]. The level of online shopping in China is trending downwards from the eastern coastal areas to the western rural mountainous areas [57].

3. Data and Method

The data in this article are from the 2015 China County E-commerce Index, which is the online business index and online shopping index of all the counties in China published on the website of the Ali Research Institute. These indexes are based on the e-commerce data of Taobao.com (accessed on 15 September 2019), China's largest e-commerce website, and are very comprehensive. The Online Retail Index (OBI) and Online Shopping Index (OSI) are constructed by Ali Research to characterize the level of e-tailing and online shopping in each county. The formulae are as follows:

$$OBI = 0.6 \times \alpha + 0.4 \times \beta \quad (1)$$

$$OSI = 0.6 \times \gamma + 0.4 \times \rho \quad (2)$$

where OBI stands for Online Sales Index and OSI represents Online Shopping Index; α stands for e-tailing density, that is, the ratio of the number of e-retailers to the population; β indicates e-tailing trade level, which is the ratio of the number of e-retailers with an annual turnover of over RMB 240,000 to the total number of e-retailers; γ stands for online shopping density, that is, the ratio of the number of online consumers to the population; ρ means online shopping consumption level, which is the ratio of online shopping consumption to the d represents the level of online shopping consumption, that is, the ratio of online shopping consumption to the number of online shoppers. The values range from 0 to 100, with larger values indicating a higher level of online shopping development. The socioeconomic data used in the regression model are partly derived from the China County Statistical Yearbook and partly crawled from Tianyancha.com (accessed on 23 April 2020). The offline retailer, ICT service providers and logistics service provider data obtained from Tianyancha.com (accessed on 23 April 2020). The length of high-grade roads, per capita GDP, employment in service industry, industrial output value above designated size, number of manufacturing enterprises, population density, and per capita savings data come from the China County Statistical Yearbook. After sorting out the data, they are all unified to the county level for analysis of the mechanism.

This paper conducts an exploratory geographic data analysis based on ArcGIS software to explore the distribution patterns of spatial data and visualize the exponential grading of online business and online shopping. We calculate simultaneously the local autocorrelation index of online business and online shopping to obtain their cold and hot spots distribution and identify their local spatial autocorrelation characteristics. The average index per hundred kilometers is calculated to verify the spatial diffusion characteristics of e-commerce. Finally, a least squares regression model is used to explore the factors influencing the diffusion of online business and online shopping.

4. Differences in Spatial Diffusion

4.1. Comparison of Spatial Diffusion Characteristics

We use the core–periphery structure based on China's "east–west" structure. Due to the large number of county units, in verifying the spatial diffusion pattern of online business and online shopping, we can make judgements based on the transition characteristics from core to peripheral counties. If the spatial diffusion from east to west conforms to the pattern of gradient diffusion from core to edge, then it is in line with the innovation diffusion hypothesis at the county scale. Conversely, there is the efficiency hypothesis. Since the cross-sectional data of each year can be seen as the result of the previous diffusion, we use the 2015 online business and online shopping data as the diffusion results of the previous years to study the spatial diffusion of e-commerce.

The distribution of county-level online business in Figure 1 shows that the high value of the index is distributed in eastern coastal China, where high-value contiguous areas are formed. The central region, as a transitional region between the eastern and western regions, has a lower index than the eastern region. Online business in the western counties is dominated by a low-value distribution. The county-level online business index has

obvious gradient distribution characteristics. High values are distributed in the eastern coastal areas from Bohai Bay to the Pearl River Delta. The mid-level is North China, Central China and South China, and the vast western area is the outermost level. The eastern region has a higher level of economic development and is also a center of technological innovation. The general forms of technology diffusion are hierarchical and contagious diffusion. The eastern region, being at the heart of online business technology innovation, is the first to be exposed to online business technology and therefore, has an advantage in online business adoption.

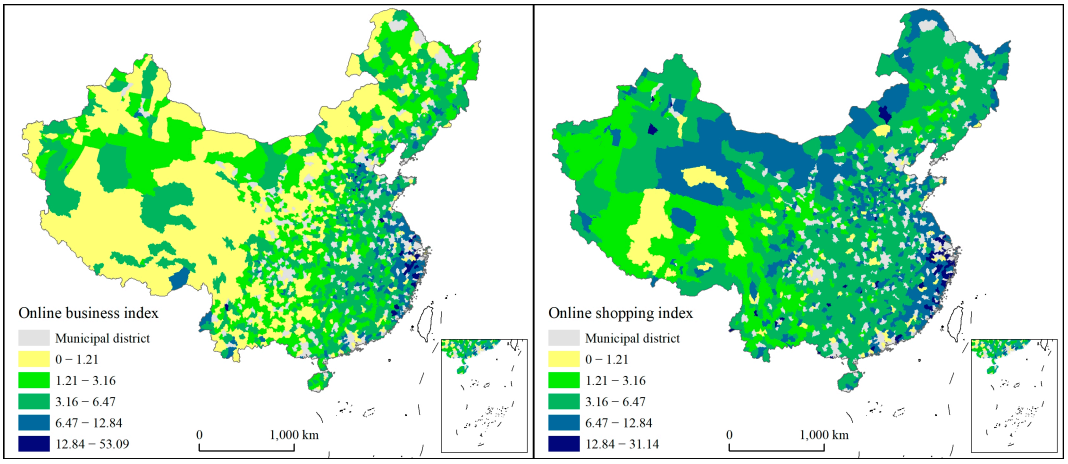


Figure 1. County-level online business and online shopping index in 2015.

The state of online shopping in counties shows that high-value areas are more widely distributed than are online businesses. High values are distributed not only in the eastern coastal areas, but also in the North China Plain and northern regions. Low values are located in the western region and are distributed over a smaller area. Comparing the spatial characteristics of online business and online shopping in counties, it can be seen that online shopping in general conforms to the same gradient distribution characteristics. However, there are many high values in the remote areas of Northeast and Northwest China, as well as on the North China Plain. The presence of these areas seems to diverge from the gradient distribution characteristics of the core–periphery structure from east to west. Compared to the eastern region, the peripheral counties have larger administrative districts but are more sparsely populated. Retail services are mainly located in administrative premises and most residents have difficulty accessing these services. The widespread distribution of internet infrastructure and high penetration of mobile internet in China, as well as the low technical threshold for online shopping, has led residents to switch from offline consumption to online shopping, with higher indices in some marginal areas.

4.2. Spatial Agglomeration Characteristics

This analysis of the distribution of spatial diffusion provides a preliminary perception of the diffusion characteristics of online business and online shopping. In the core and peripheral areas, we find some high values that presumably reflect agglomeration, but whether these areas are truly high-value clusters still needs to be explored using more scientific spatial analysis. In Figure 2, hot-spot analysis is used to explore the high-value agglomeration of online business and online shopping. The analysis of the online business index shows that the hot spots are concentrated in the eastern part from Bohai Bay to south-western Fujian, with a significance level of more than 90%. Most of China's top 100 counties in terms of GDP are located in the eastern region, which have a good manufacturing base, a complete retail supply chain and abundant human capital. As a result, the high value areas for online business are clustered in the eastern regions. Because of the mixed distribution of high and low values in North China, Central China, and South China, the analysis results

are not significant. Most of the areas in the west are cold-spot clusters with a significance level of more than 90%, supplemented by the presence of some small patches of mixed high and low values in the west, which show non-significant results in the analysis of cold and hot spots. The hot-spot analysis shows the characteristics of the gradient distribution of the online business index more clearly, but there is a non-significant area in the northwest region, and further demonstration is needed.

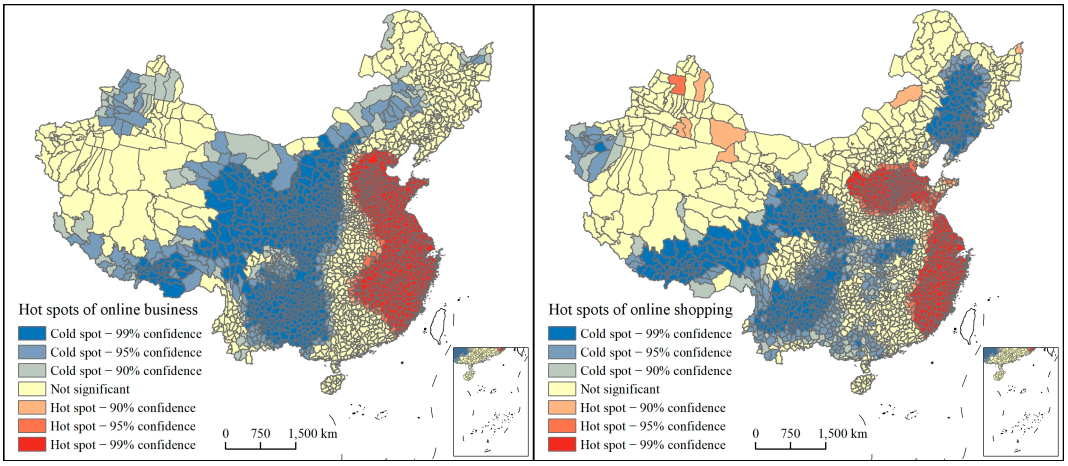


Figure 2. County-level hot spots of online business and online shopping indexes in 2015.

The online shopping index has formed a very large hot spot in the eastern region and its hinterland, North China. We observe only that there are high values in North China in the previous analysis, but we do not know whether this area represents a "marginal reversal". The analysis here suggests that the high-value area in North China is not a "marginal reversal" but is part of the greater whole of the eastern core area. However, this result also needs further verification. Additionally, there are six small hot spots in the northern region. Whether there is a "revolution" in the peripheral regions also needs further investigation. The northern fringe regions are mostly mountainous, grassland or desert terrain, with poor accessibility to retail services and a strong incentive for residents to shop online. Moreover, the industries in the northern regions are mostly resource extraction and animal husbandry, where the inhabitants have a high per capita income and a strong spending power, thus creating high-value agglomerations.

4.3. Differences in the Same County

Having analyzed the spatial diffusion patterns of online shopping and online business indexes in different counties, we will explore which pattern is more dominant within the same county. Especially in remote regions, if one of these indexes has a greater advantage and is widely distributed, this can indicate the existence of marginal reversal. Therefore, the difference between the online business and online shopping indexes is used to represent the divide in Figure 3. Online shopping has an absolute advantage in the northern region, including the fringe areas from the northeast to the northwest, which form an agglomeration. There are few regions in eastern China, where online shopping has an absolute advantage except for the region on the west coast of the Taiwan Straits. This finding shows that peripheral areas have a demand for technology but are better adapted to online shopping and less well adapted to online business. Combined with the previous analysis, it can be seen that the eastern coastal region takes into account the application of business-oriented and consumption-oriented technology, which is a relatively comprehensive technical adaptation. Marginal areas are subject to regional endowments, industrial foundations and knowledge thresholds, resulting in poorer application of business-oriented technology, while the application of consumption-oriented technology has an advantage.

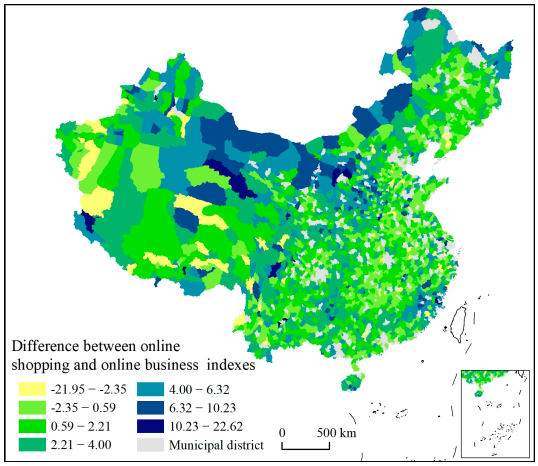


Figure 3. County-level differences in online business and online shopping indexes in 2015.

4.4. Diffusion Features in the Core-Periphery Structure

From the perspective of spatial distribution, online business and online shopping have different diffusion features, and it can be concluded that there are differences in their spatial diffusion patterns. To examine this result in more depth, the eastern coast (from Bohai Bay to the Pearl River Delta) is selected as the baseline, and the average values of online business and online shopping indexes within a radius of 100 km from the baseline are calculated. At the same time, the three largest cities in China—Beijing, Shanghai, and Guangzhou—are selected as benchmark points for cross-comparison with the baseline. The eastern region, as the center of innovation for ICT and e-commerce in China, is the starting point for the proliferation of e-commerce. Therefore, we have chosen these two sets of baselines and benchmark points.

According to the average value per 100 km in Figure 4, the spatial spread of online business basically shows a decreasing trend from the core area to the edge. There are only two peaks around A (2500 km) and B (3200 km), but neither of them is sufficiently sustained and robust. On the whole, online shopping presents characteristics of decreasing from east to west, but there are some regions where this decreasing pattern is broken. Two crests appear in regions C (1200–1800 km) and D (2300–3200 km), both of which remain in a longer range and exhibit good robustness.

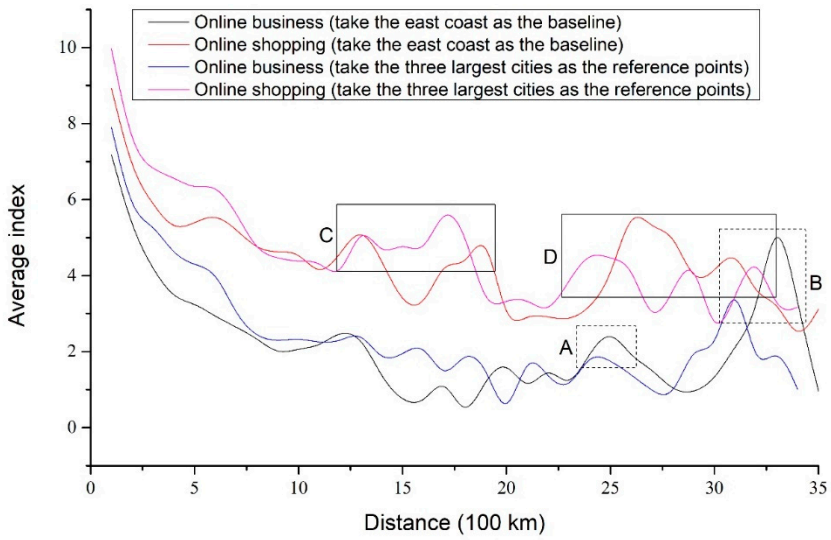


Figure 4. Average index of online business and online shopping decaying with distance. A, B, C and D are crests of online business and online shopping.

Summarizing all the diffusion characteristics analyzed above, we can affirm that online business basically decreases in a gradient from the core area to the peripheral area, which is in line only with the innovation diffusion hypothesis. In addition to the diffusion of innovation from high-tech areas to low-tech areas, the phenomenon of low-tech areas preferentially using technology in response to their own needs also exists in online shopping, generally conforming to the innovation diffusion hypothesis, and some areas also conform to the efficiency hypothesis. Thus, the analysis in this section essentially verifies the previous analysis.

5. Why Are There Differences?

As the previous analysis shows that the spatial diffusion patterns of online shopping and online business differ, regression methods are used to explore the reasons. We take the county-scale online business index and online shopping index as dependent variables to explore the mechanism of the spatial diffusion difference. As Coe and Yeung noted, many social and cultural aspects of e-commerce development are embedded in specific places and environments [63]. Innovative applications of e-commerce cannot appear in a vacuum but need to combine sufficient internet infrastructure, human capital, entrepreneurship, and supportive finances, logistics and government agencies. Boschma and Weltevreden also found that the adoption of the internet is affected by the unique characteristics of enterprises, network relationships and geographic locations [64]. When selecting elements that affect the adoption of online business, the possibility of offline retail transforming into online business is considered, and infrastructure elements, such as ICT services, logistics services, and the length of high-grade roads, are also included [65]. Areas with good infrastructure are in a better environment to adopt online business and are more likely to adopt new technology [66]. In terms of the offline retail, the socioeconomic environment, per capita GDP, employment in the service industry, the output value of industrial enterprises above a designated size, and manufacturing enterprises are selected as indicators. The greater the number of offline retailers, the better the entrepreneurial climate in the region. Previous research has found that a significant amount of online business has been transformed from offline retail [67]. Economic growth accompanied by a good institutional environment and highly skilled human capital have a catalytic effect on the use of online business [68]. A quality labor market can reduce labor costs for online business companies, and the matching of labor markets with companies is smoother [69]. The higher the output value of industrial enterprises above the scale and manufacturing enterprises, the richer the variety of products in the region, supporting online business development in terms of production in the supply chain [70].

As Farag et al. stated, shopping accessibility, that is, the number of stores within a fixed region, is an important variable affecting residents' online shopping [2]. Cao et al. incorporated internet infrastructure, education level, and income into the regression model to investigate the spread and use of online shopping, and the results were also significant [54]. Clarke et al. suggested that age and income are crucial demographic discriminators of e-commerce usage [6]. Since the number of offline retail options reflects local shopping demand to a certain extent, the number of offline retail options is used as a variable. Infrastructure elements, such as ICT services, logistics services, and the density of high-grade roads, are also considered independent variables, similar to online business. The total per capita savings and per capita GDP that affect residents' consumption decisions are also selected as an independent variable. These two factors are related to spending power. The higher the GDP per capita and total savings per capita, the higher the likelihood of online consumption [71]. Counties with low population density indicate remote areas, and the population density variable is added to verify whether remote areas have higher consumer demand. Descriptive statistics are provided in Table 1.

Table 1. Descriptive statistics.

Variable	Min.	Max.	Mean	Std. Dev.
Offline retailer	27	80,913	6190.84	6007.36
ICT service providers	2	48,051	378.67	1293.33
Logistics service providers	1	2018	61.85	64.49
Length of high-grade roads	0	1510	200.17	205.48
Per capita GDP	12.95	17,023.24	1173.90	1116.26
Employment in service industry	68	1,407,234	69,589.93	74,135.89
Industrial output value above designated size	0	7852.39	254.07	463.22
Number of manufacturing enterprises	7	59,024	2692.67	5032.81
Population density	0.12	2062.5	295.41	282.15
Per capita savings	316.8	156,913	19,832.61	14,248.6

Stata was used to perform OLS regression analysis. Table 2 shows that the fitting coefficients of the model are all above 0.45, and the model is in line with expectations. In the regression model of online business, the amount of offline retail has a significant positive effect on the development of online business. As a relatively new retail form in the Internet era, online business is also rooted in and transformed from offline retail. Strong offline retailing is a necessary foundation for the development of online business [72]. In terms of infrastructure, the number of information technology service providers and the density of high-grade roads have a significant positive impact on the development of online business, while the regression coefficient of logistics service providers is not significant. The development of online business is inseparable from the growth of the internet, logistics, and transportation. Infrastructure is an essential condition for the growth of online business [73]. In the macrosocial environment dimension, except for the industrial output value above a designated amount, other variables, including per capita GDP, service industry population, and number of manufacturing companies, have significant positive effects. These factors are indispensable for the development of online business; in particular, the local manufacturing industry can provide the necessary commodity supply chain for the development of online retail, and the human capital of the service industry can provide relevant technical labor and promote the spatial diffusion of online business [63].

Table 2. Ordinary least squares (OLS) estimation.

	Online Business	Online Shopping
Constant	1.512 ***	1.711 ***
Offline retailer	7.53×10^{-5} ***	4.24×10^{-5} ***
Telecom service providers	0.000 **	0.000 ***
Logistics service providers	0.001	0.003 ***
Length of high-grade roads	0.001 **	0.001 ***
Per capita GDP	0.000 ***	0.000 ***
Employment in service industry	1.49×10^{-6}	
Industrial output value above designated size	-7.19×10^{-8} ***	
Number of manufacturing enterprises	0.000 ***	
Population density		0.002 ***
Per capita savings		0.000 ***
R2	0.467	0.59
Ad R2	0.464	0.589
Observations	1919	1919

Significance at the 10%, 5%, and 1% levels is indicated by *, **, and ***, respectively.

In the online shopping regression model, the number of offline retailers in the region has a positive effect on online shopping. It is generally believed that the demand for online shopping in remote areas is due to the lack of offline retail services, which seems to contradict our results. In fact, the demand for online shopping does not increase without a foundation. Most online shopping is transformed from offline demand, and the growth

of online shopping also develops from the offline economy. However, online shopping is also restricted and affected by its own economic capabilities, whether in core or remote areas. Regions with higher per capita GDP and per capita total savings have higher demand for online shopping [54]. The results show that the regression coefficients of per capita GDP and per capita savings are both significant positive effects, which supports the previous research conclusions. Among the infrastructure elements, the regression results of information technology services, logistics services and the length of high-grade roads are significant positive effects, which are important supporting conditions for promoting the development of online shopping. The regression results show that population density has a positive effect on online shopping. This result is different from the existing assumptions that the lower the population density is, the greater the demand for online shopping in remote areas will be. This discrepancy may be because the demand for online shopping in areas with low population density is also bounded by a composite of affordability and infrastructure.

6. Discussion

With the advancement of ICT, there is ongoing debate about its impact on core and peripheral areas. Especially with the increasing popularity of e-commerce, research on the spatial diffusion mode of e-commerce has been the focus of geographical research. The main focus has been on the spatial diffusion model of online shopping in different regions. Over the past few decades, marginal areas, including rural areas, have undergone a process of de-agriculturalization and post-productivism, gradually transitioning from an agricultural economy to more diversified economic activities, including handicrafts, retail and tourism [74,75]. However, scholars have still conducted little research on online shopping in peripheral areas and have paid more attention to the spatial diffusion mode of online business and its impact on peripheral areas. Online business in rural China has developed rapidly with the support of e-commerce platforms and the government. In particular, the development of Taobao Village has resulted in the accumulation of much experience and seems to be a "marginal revolution" realized through internet technology in remote areas [76,77].

From a regional perspective, we have clarified the differences in the diffusion characteristics of core and peripheral regions through a comparative study of online business and online shopping diffusion in China. At the same time, we have a clearer understanding of the different factors that influence spatial diffusion. The diffusion of online business in Chinese counties is in line with the hypothesis of innovation diffusion. The dominant area for the development of online business is the eastern region, which not only has a relatively complete network infrastructure but also benefits from a strong industrial foundation that provides the necessary supply chain for the development of online business. The efficiency hypothesis of online business is not yet supported in peripheral areas, and the effect of using internet technology to overcome spatial friction is not obvious in these areas. County-level online shopping conforms to the efficiency hypothesis, which has been confirmed in previous "urban–rural" comparative studies, but overall, online shopping is in line with the innovation diffusion hypothesis. Rural and peripheral areas have the motivation to adopt online shopping, but its transformation into real shopping behavior is restricted by the level of income and savings. Affordability is, therefore, the main reason why some marginal areas have achieved a "marginal revolution" in terms of online shopping.

Online business was once considered to be a momentous way to narrow the gap between the peripheral and core regions, especially at small scales, and such findings can often be obtained [60]. From the findings of this paper's country perspective, it is clear that there is a large gap between online business development in peripheral regions and core regions, and that online business development is still higher in core regions with first-mover advantages. The development of online business has widened the regional gap to a greater extent [61]. Due to differences in regional endowments and industrial bases, industries in peripheral regions have difficulty supporting retail development. There may be a few

exceptional cases in marginal areas of high online business indices, but at the national level, this assumption of trying to rely on online business to narrow regional economic differences is difficult to achieve. Online shopping is more capable of bridging core and peripheral differences than online business, a capability that refers not to the ability to bridge economic differences, but to the potential to reduce spatial consumption inequality and the welfare gains [71]. As a new trade and distribution technology, online shopping can increase urban–rural and inter-regional trade and alleviate spatial consumption inequalities. The marginal areas may not have the industrial base to support the development of online business, but the reverse gradient diffusion of online shopping shows the possibility of increased accessibility of shopping for the residents of the marginal areas, and online shopping shows the potential to significantly improve the convenience and quality of life of the residents of the marginal areas.

We acknowledge that the development of online business will lead to the growth of some marginal and rural areas, and there are many success stories in China, Europe, Africa and South Asia [52,78–80]. However, on a national scale, it is clearly unrealistic to try to rely on online business as a means of reducing regional economic disparities. What needs to be known is that not every village is suitable for online business development and that villages in core regions will always have a comparative advantage in developing online business. As a local development policy, therefore, online business is a better way to achieve regional economic revitalization. However, online business is not suitable as a policy to balance regional differences at the national level for either developing or developed countries. In European countries in particular, the economic function of the countryside is not particularly emphasized, and the countryside is used more as a natural, relaxing and comfortable place, where its economic function is not significantly underlined [81,82]. Therefore, we are keen to make people aware of the considerable role of online shopping in promoting consumption equality through our research. The development of online shopping to improve the accessibility of shopping for people living in marginal and rural areas and to reduce regional consumption inequalities is the greatest benefit that online shopping gives to marginal areas. Online shopping is a significant tool for improving regional inequalities that applies equally to developed and developing countries.

Given the complexity of the various systems of e-commerce, certain issues have not yet been considered. Spatial reconstruction has been brought about by changes in technology, with counties in different regions being drawn into the digital economy to varying degrees. Research has focused more on large cities and comparatively ignored non-metropolitan areas and rural areas. Therefore, it is especially important to prioritize the analysis of the spatial diffusion characteristics of e-commerce in different areas. Research on the spatial diffusion of e-commerce is concerned not only with its impact on the spatial structure of the economy but also with its impact on changes in the spatial structure of cities and regions. Will e-commerce expand the agglomeration advantages of cities and other core areas, or will it weaken the comparative advantages of cities to bring diverse development opportunities to peripheral areas? These issues are worth discussing in the context of internet research.

7. Conclusions and Policy Insights

Retail and consumption in e-commerce are distinguished to compare the spatial diffusion patterns of online business and online shopping at the county level. This article also attempts to determine which hypotheses are fitted by online business and online shopping in the core–periphery structure at a large scale and to explore the mechanisms behind these differences in diffusion modes.

Our comparative study found that eastern China has an absolute advantage in online business. The spatial diffusion of county-level online business obeys the characteristics of a gradient distribution from east to west. Gradient transition characteristics are also the main diffusion characteristics of online shopping, but there are many continuous areas with high values in the remote areas of the northeast and northwest and the North China

Plain, breaking the pattern of gradient distribution. Online business exhibits a hierarchical diffusion of technology only from strongly urbanized areas to weakly urbanized areas, which conforms only to the innovation diffusion hypothesis. In addition to online shopping conforming to the innovation diffusion hypothesis as a whole, there is an edge rise in low-tech areas, due to their own need to actively adapt to technology. The efficiency hypothesis is also supported in peripheral areas between 1200–1800 km and 2300–3200 km from the core. Online shopping has an advantage in some remote areas in the west. However, most of the coastal areas have experienced comprehensive technical adaptation, taking into account both business and consumption technology applications.

The difference in spatial diffusion between online business and online shopping at the county level mainly exists because they are different types of internet applications. Online business is a retail-oriented internet application that requires not only internet infrastructure and user skills but also the support of macrosocial economic conditions, including the accumulation of economic factors, such as an industrial foundation and human capital. For the spatial diffusion of online business, it is not enough for residents to have internet skills alone; the online business also needs to be integrated with industry, and without the support of an industrial base, it is difficult to realize the efficiency hypothesis at scale in marginal counties. Therefore, the socioeconomic gradient between the east and west in China determines the characteristics of online business, and the structure of online business in China conforms only to the innovation diffusion hypothesis at a large scale.

Online shopping differs from online business in that it is a consumption-oriented internet application and has fewer restrictions. The role of income and savings is particularly important for online shopping, as the mobile internet infrastructure and its use are no longer issues in urban and rural areas of China. Residents' online shopping is restricted by their own budgets, and other factors have little impact. Although in the overall core–periphery structure, online shopping is associated with the innovation diffusion hypothesis, this does not prevent online shopping from achieving a breakthrough in the diffusion pattern in peripheral areas. In China, where the mobile internet is well developed, infrastructure elements and skills are no longer obstacles to online shopping. Residents in the west with high incomes and savings can easily shift their consumption patterns to overcome spatial friction.

The article also has some limitations. Time series data can provide better insight into the spatial diffusion patterns of e-commerce, but the data used in this article are not rich enough, due to the limited availability of data. In the future, it will be possible to obtain richer data and study the latest proliferation patterns of e-commerce. In addition, the data used in the regression model in this paper are cross-sectional data, and the results of the analysis are correlational rather than causal. We will look further into the panel data for regression analysis to explore the mechanism of e-commerce diffusion.

Some policy insights can be drawn from this study. For policy makers, while it is impractical to rely on online business to achieve a widespread rise of marginal areas, policy releases still need to be more tilted toward marginal areas, including the improvement of infrastructures, such as the internet, transportation and logistics, as well as the support of fiscal and taxation systems, such as microfinance and tax incentives. In particular, the layout of the infrastructure reduces the inequality of public services between the core and the periphery, bringing opportunities for the development of online business in the periphery and contributing to the equalization of consumption between the core and the periphery. At the same time, other actors, such as social groups and non-profit organizations, need to pay more attention to the ability to use digital technology in marginal areas outside of cities and contribute more to raising the level of online business in marginal areas. In addition, social groups need to create more ties for the development and exchange of e-commerce between urban and rural areas in order to facilitate integration between core and peripheral areas.

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